What we're doing

For the registered default contacts-v1-exp277-m2-p06-full-epoch-1.5B checkpoint at step 266344: (1) how many true contacts, or what fraction, put generation on the right path, and does coverage of secondary-structure elements help; (2) how does eval-val R-precision vary with length; (3) on verified AFDB training entries contradicted by subsequently released experimental structures, does MarinFold favor the AFDB contacts or the experimental contacts?

Why

A few distributed correct contacts may disambiguate the fold more efficiently than uniformly sampled contacts. Predictions on training-label errors may reveal whether the model generalizes beyond its AFDB supervision.

Accuracy versus length

All 97 eval-val proteins; exp277 step 266344, canonical 100-rollout scores. All-range R-precision: 0.554 (95% CI 0.512–0.594). Long-range R-precision: 0.538 (95% CI 0.494–0.580). Length association: Spearman rho −0.068, p=0.507 (all-range). No detectable monotonic trend in this sample; longest bin has only eight proteins.

Contact injection: a useful nudge, no sharp transition

All 6,664 conditions / 666,400 rollouts complete on the 97 eval-val proteins. Eight random contacts: remaining R-precision 0.610; gain 0.061 (95% CI 0.046–0.079). 5% target budget: remaining R-precision 0.615; gain 0.064 (95% CI 0.050–0.081). Both policies first cross the predeclared five-point mean-gain criterion at eight contacts / 5%. Random peaks at 0.711 (95% CI 0.698–0.724), at the 40% budget. No tested dose reaches mean R-precision 0.80. Long-range results are similar. Supplied pairs never score. As dose rises, the remaining truth and R shrink; the declining absolute curve at large budgets does not establish that more context harms predictions.

Secondary-structure coverage does not improve the early resp

One contact per coverable element uses a mean 16.9 contacts / 7.04% of truth. Remaining R-precision: 0.628 balanced versus 0.626 random at the same counts. Paired difference 0.0023 (95% CI −0.0051–0.0098): no clear advantage. Two contacts per element also show no clear advantage. At 40% truth supplied, random exceeds balanced by 0.0187 (95% CI 0.0119–0.0253). The policy comparison also changes contact composition: one balanced pass contains fewer long-range contacts than random sampling at the same count.

Completeness and robustness

Zero token-cap hits across all 666,400 rollouts; 2,218 invalid rollouts (0.333%). All samples retained, with malformed / nonexistent-residue pairs unable to vote. Zeroing entire invalid rollouts changes all-range curve means by less than 0.00022. Fresh unconditioned mean 0.55449 reproduces archived 0.55375; paired difference CI includes zero. Eight protocol tests and twenty exact canonical-scorer comparisons passed. Raw rollouts, vote matrices, timings, frozen inputs, and source code are published with checksums.

Verified AFDB/PDB disagreements

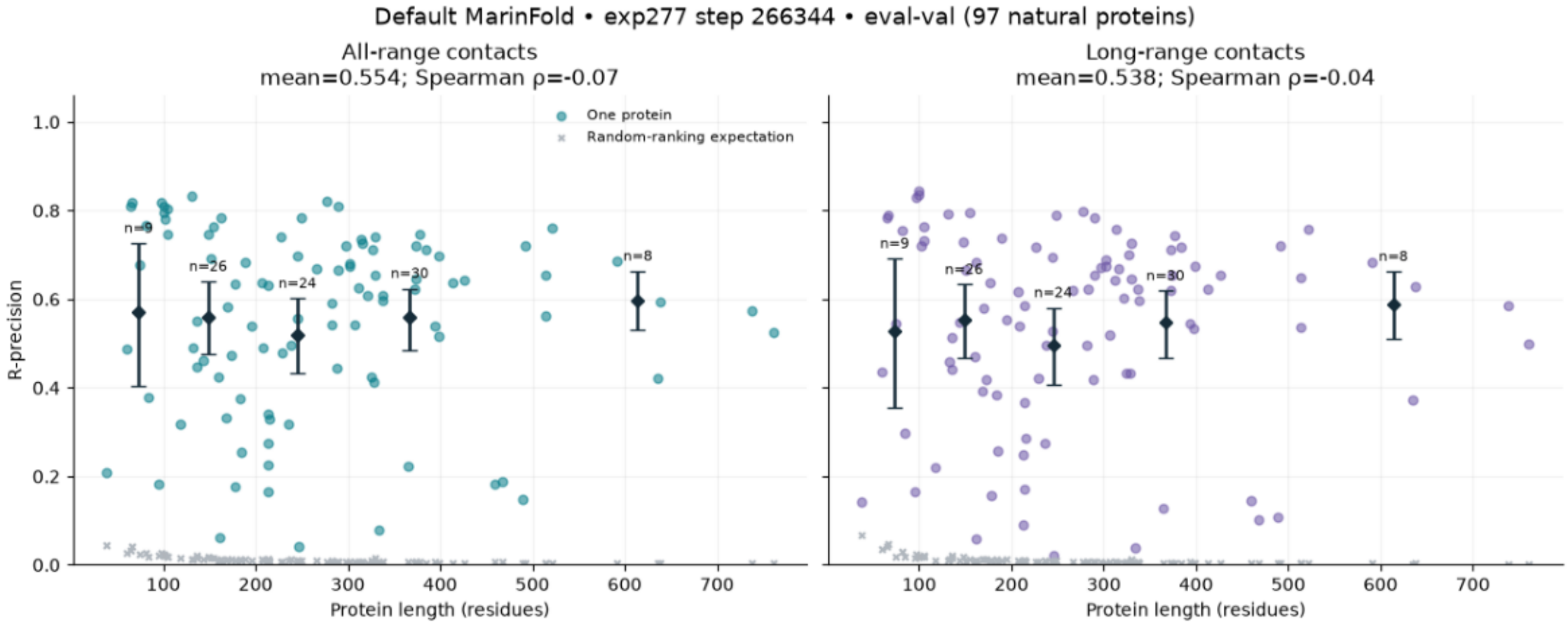
Screened 700 retained training entries; 357 usable mappings. Confirmed 20 against the original AFDB v4 source; training contacts reproduce exactly on the common resolved residues used in each comparison. Six cases selected: three large assembly-bound discrepancies (TM-score 0.25–0.33), NDUFA4 contact disagreement, and two alternate-state controls. Later PDB release dates follow the original November 2022 AFDB objects. Assembly context and biological state prevent labeling all disagreements as unequivocal folding errors.

Model preference: mixed, no reliable correction

Dynein regulatory complex subunit 2 favors AFDB: exclusive mean votes 0.206 vs 0.025; R-precision 0.326 vs 0.194. Tex36 weakly favors PDB, but accuracy is low: 0.093 versus AFDB, 0.156 versus PDB. Centrosomal protein: R-precision zero against both structures. SemiSWEET strongly favors the AFDB state; NDUFA4 is weak and OCP is mixed. All 1,200 rollouts finished. Six had generation errors; zeroing them preserves these conclusions.

accuracy_vs_length.png

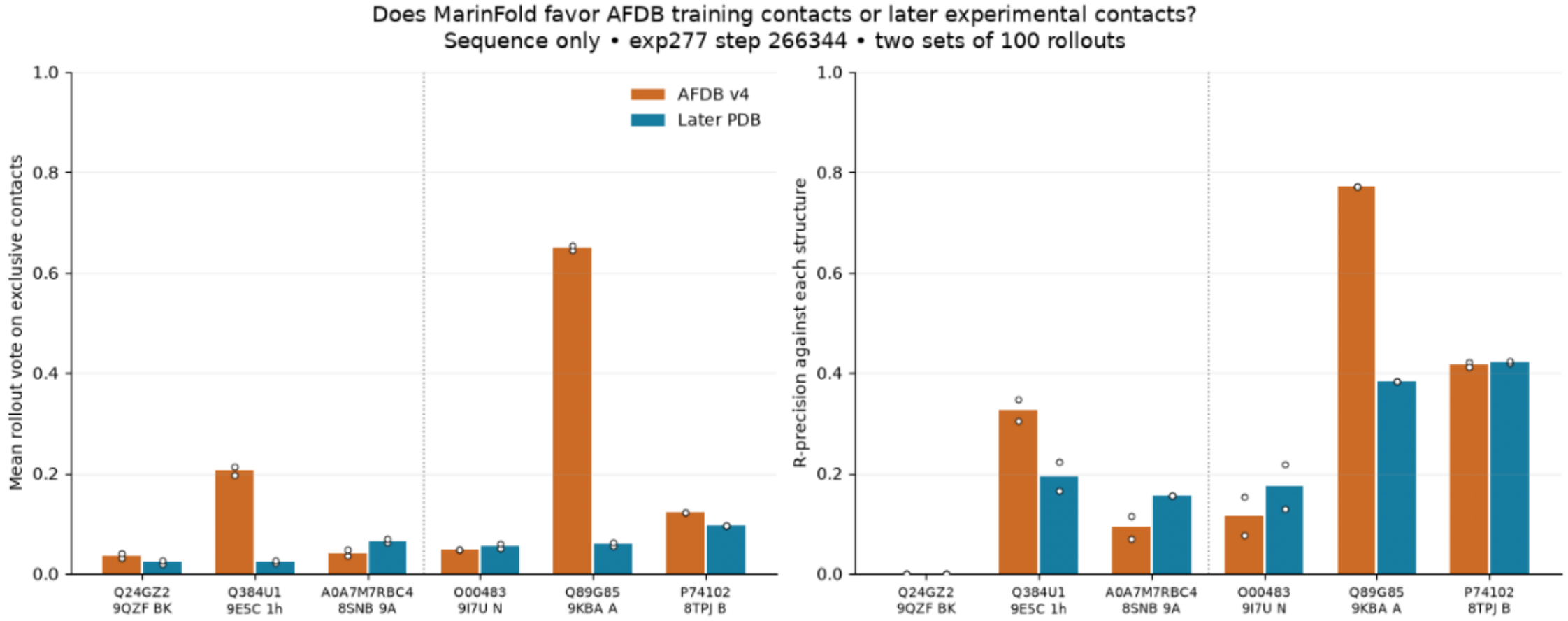
100-rollout consensus; diamonds are length-bin protein means with 95% bootstrap intervals. All: sequence separation ≥6; long: ≥24.
No new inference; no eval-test scores.



$ plot_length.py

afdb_marinfold_preference.png

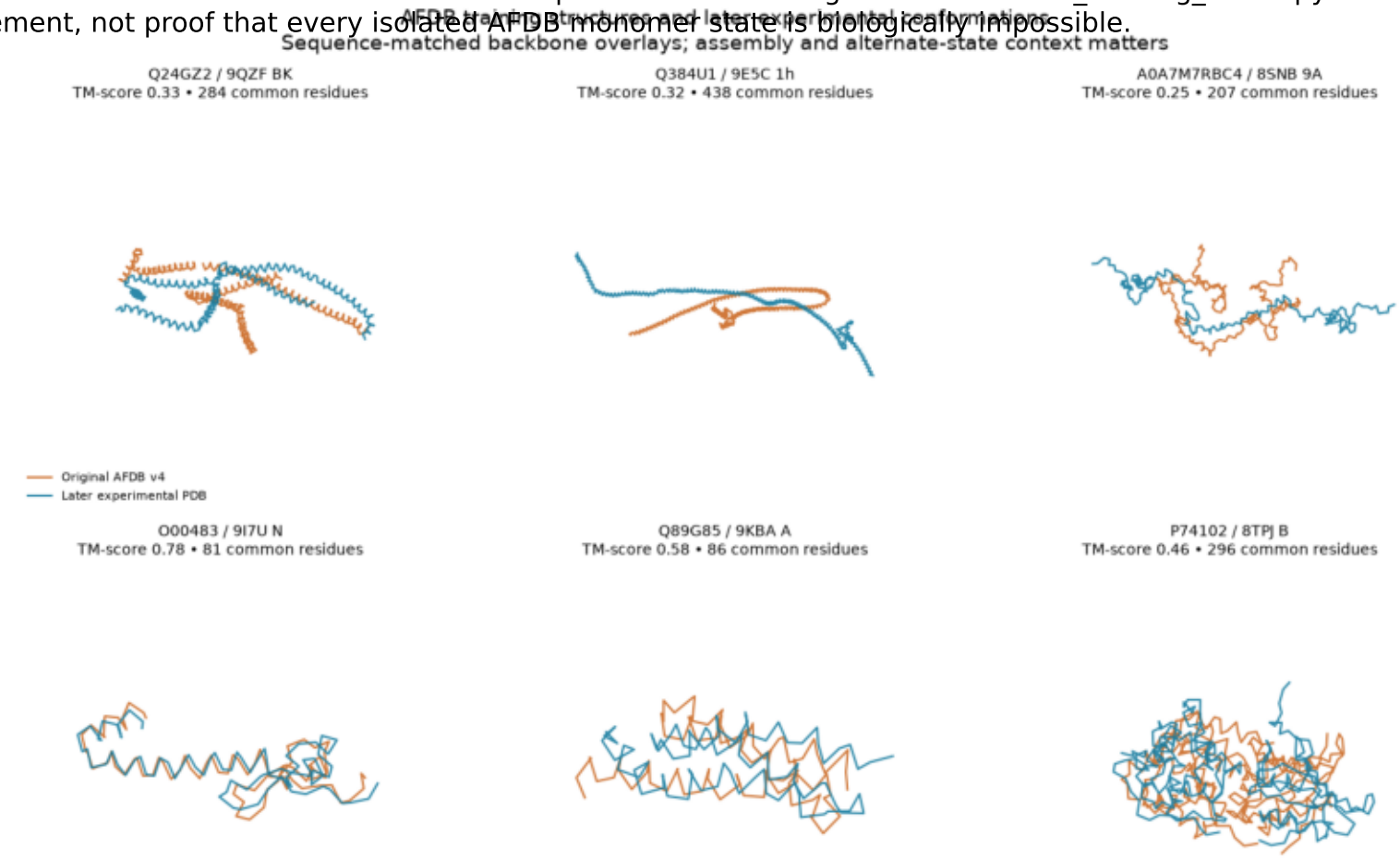
Bars average two 100-rollout repeats; dots show each repeat. Left: mean support over contacts unique to each structure, on common resolved residues. Right: R-precision against each contact set, whose sizes differ. The first three cases have large assembly-bound discrepancies; the others are contact/state controls. Predictions have no structure contacts in their prompts.




```
$ analyze_cases.py --run /data/exp369/cases-v2
```

afdb_pdb_backbones.png

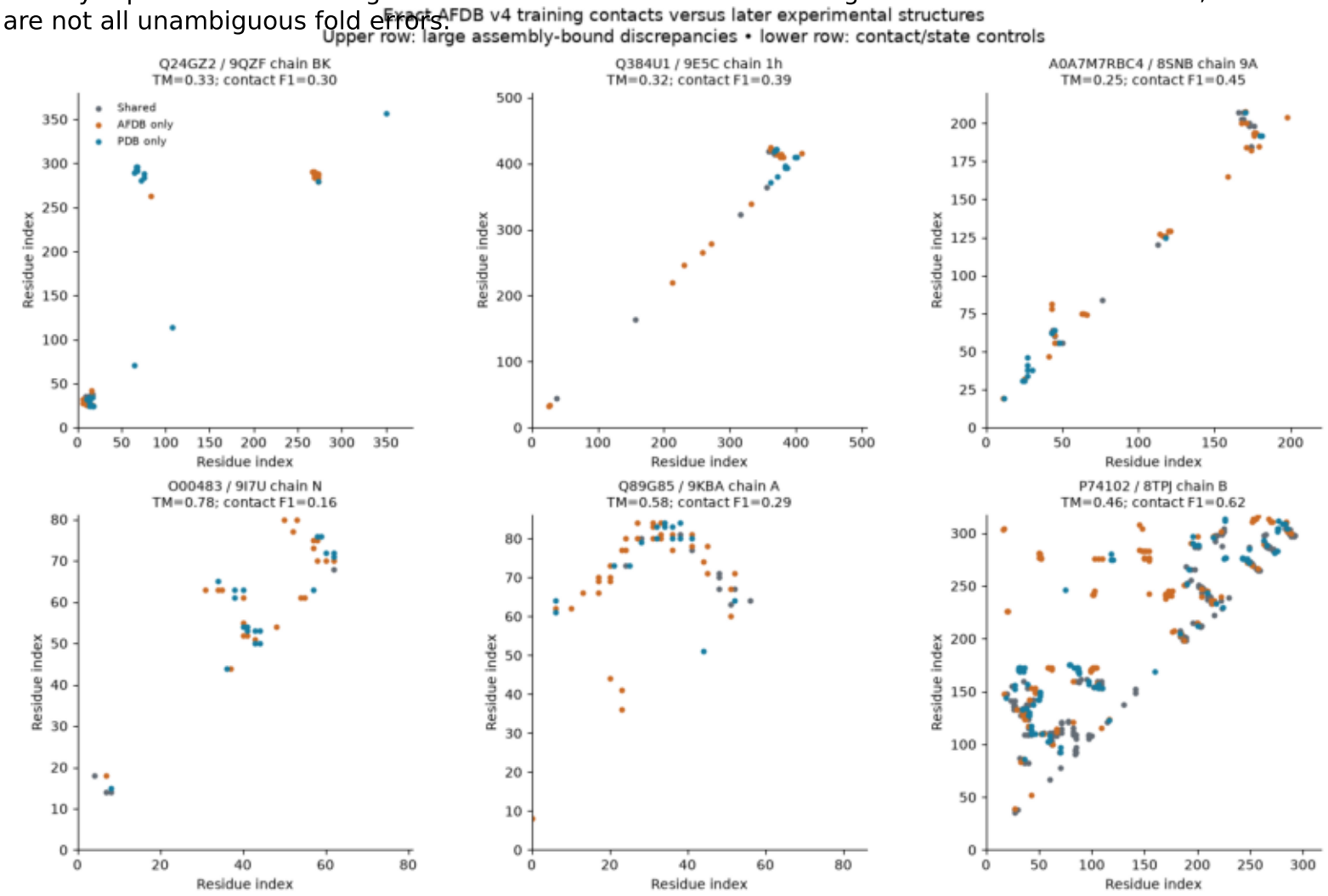
C-alpha traces are least-squares aligned using identical common sequence positions; gaps are not connected. Plot panels have independent physical scales. TM-scores in titles come from the separate structural alignment in confirm_training_errors.py. These show geometric disagreement, not proof that every isolated AFDB monomer state is biologically impossible.




```
$ plot_structures.py
```

afdb_pdb_disagreements.png

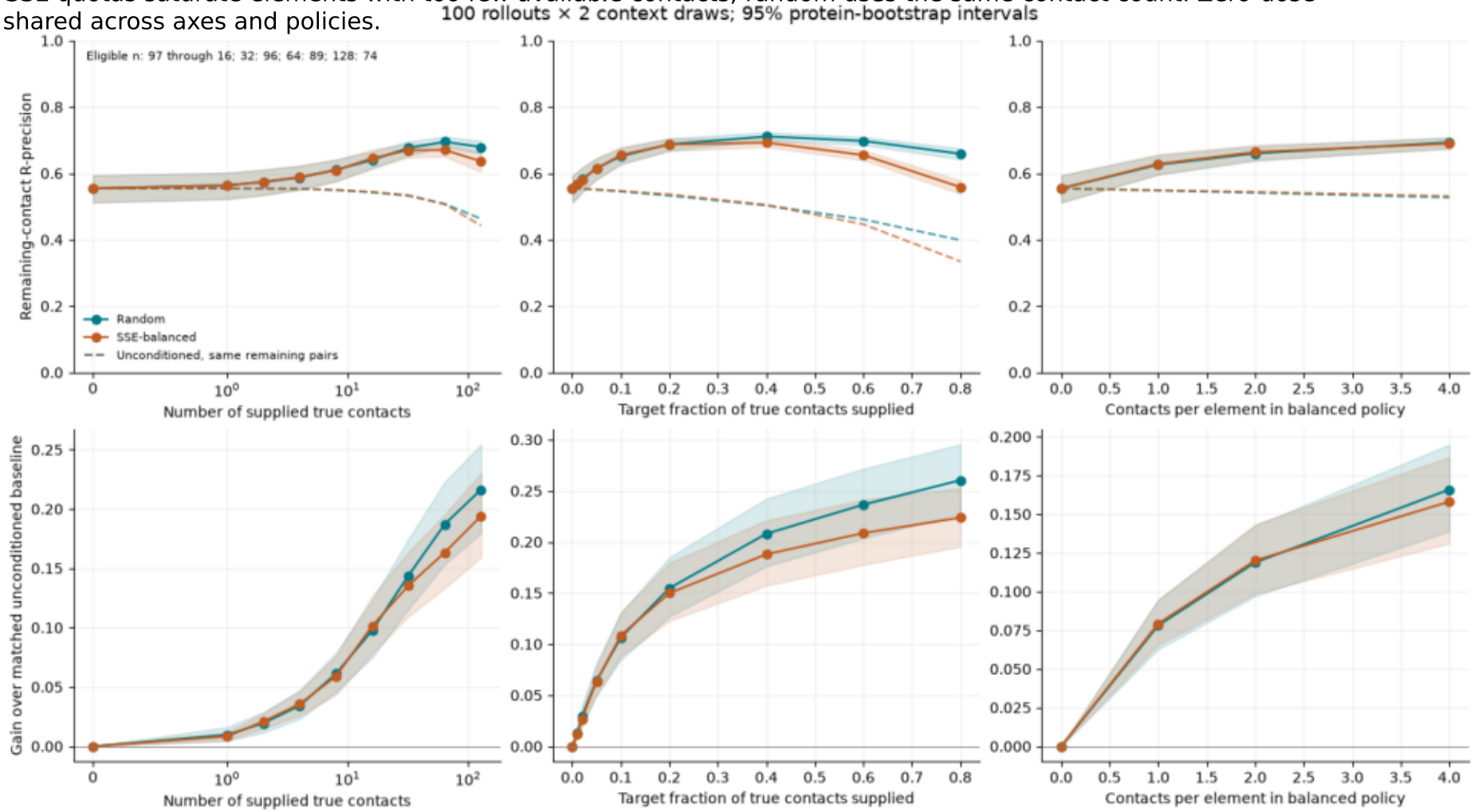
No MarinFold predictions yet. Contacts use the canonical pyconfind ≥0.001 and sequence-separation ≥6 definition on common resolved residues. AFDB contacts exactly reproduce the training documents. TM-score is structure-aligned over common residues; these assembly/state changes are not all unambiguous fold errors.




```
$ prepare_cases.py
```

spiking_all.png

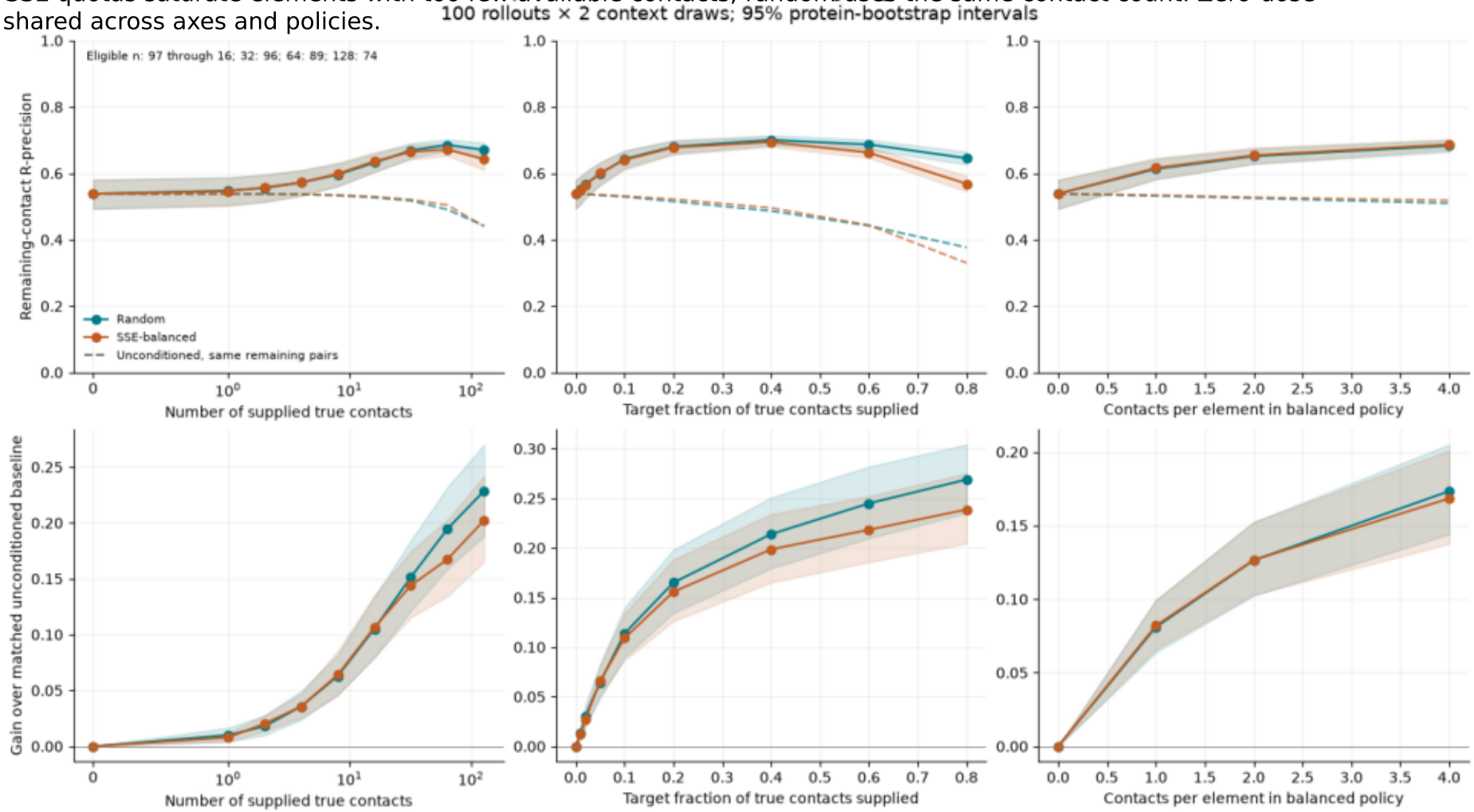
Supplied pairs never score. Dashed curves: unconditioned scores on the identical remaining pair universe. Count doses omit proteins with no truth remaining; see CSV denominators. Fraction budgets floor to integer contacts, with a minimum of one for nonzero doses. SSE quotas saturate elements with too few available contacts; random uses the same contact count. Zero-dose predictions are shared across axes and policies.




```
$ analyze_spiking.py --run /data/exp369/full-v2
```

spiking_long.png

Supplied pairs never score. Dashed curves: unconditioned scores on the identical remaining pair universe. Count doses omit proteins with no truth remaining; see CSV denominators. Fraction budgets floor to integer contacts, with a minimum of one for nonzero doses. SSE quotas saturate elements with too few available contacts; random uses the same contact count. Zero-dose predictions are shared across axes and policies.




```
$ analyze_spiking.py --run /data/exp369/full-v2
```